

Required preparation Lab 5

Similar to Lab 4, we will create a custom input system for Unity using an ESP32, however, this time, we will be using WebSockets

Learning outcomes:

- Familiarise yourself with the [Websocket Protocol](#)
- Create bidirectional wireless communication using sensors and actuators
- Learn how to create a Websocket Server using Python
- Learn how to send and receive Websocket messages to Unity using Websockets

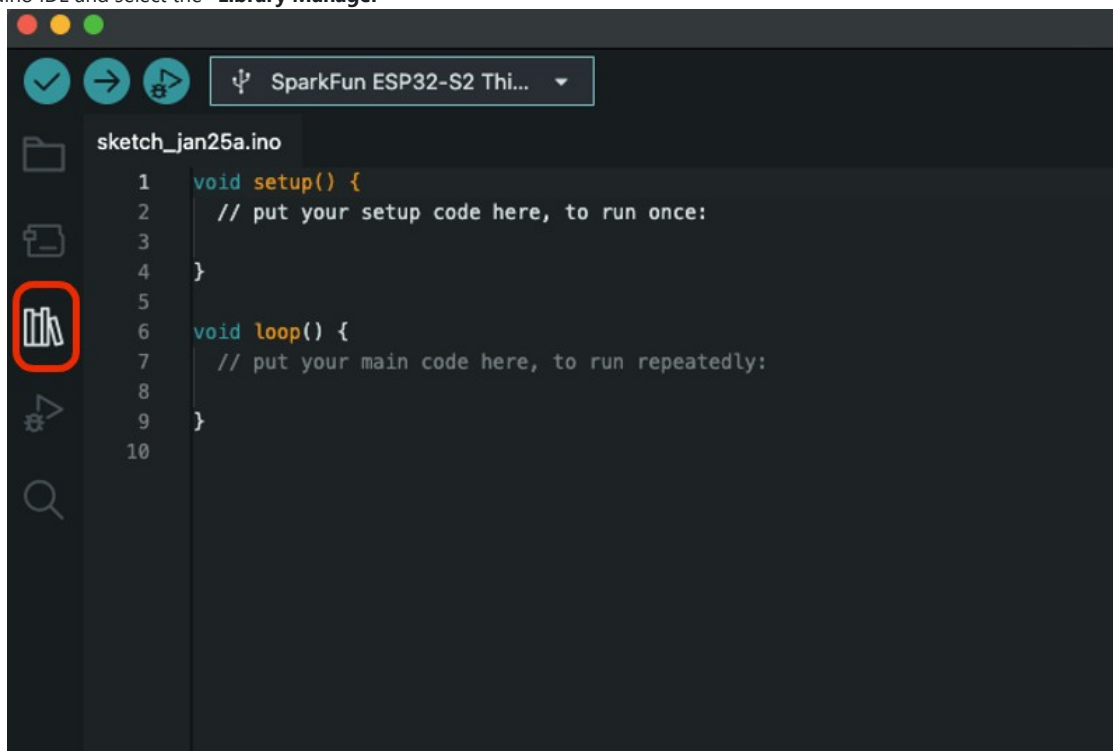
Preparation

Download in your laptop the repository for tangible interaction at ExtralityLab:

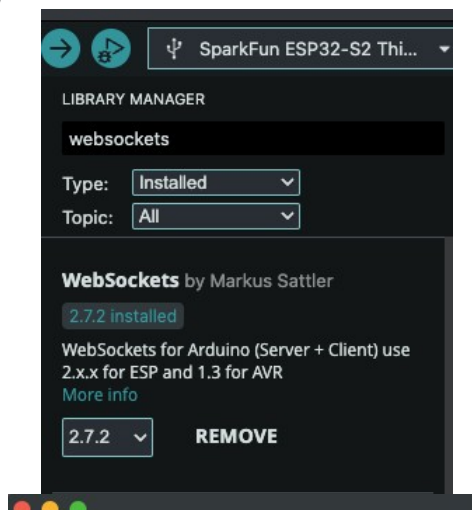
- <https://gitea.dsv.su.se/ExtralityLab/tangible-interaction>

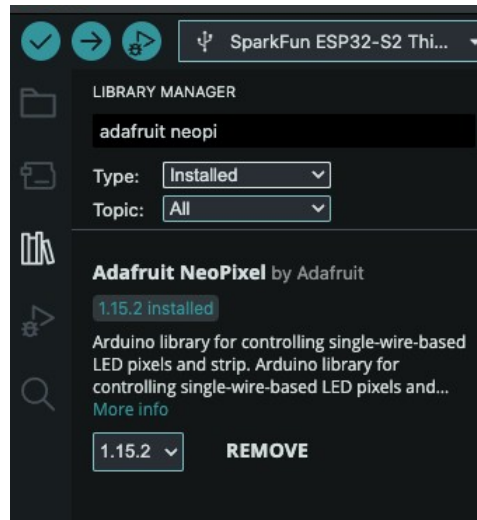
Install the Websockets and the Adafruit Neopixel Libraries on ArduinoIDE:

- Open Arduino IDE and select the "Library Manager"



- Search for the "Adafruit Neopixel" and the "WebSockets" by *Markus Sattler* libraries and install them (note you may have a more recent version than in the screenshot and that's fine)





Download and install Python 3 along with the Websockets library:

- <https://www.python.org/downloads/>

1. Navigate to the directory where you have downloaded the Tangible Interaction Gitea package and open a terminal/command line window there (e.g. Desktop/Desktop/tangible-interaction/Websockets-Communication)
2. Install the web sockets library on Python by opening the terminal and pasting the following command:

- **"pip install websockets"**
- navigate to the **"websocket_server"** folder by running **"cd websocket_server"**
- Try running the command **"python WebsocketServer.py"**. If no errors show up then you have installed everything correctly

Note: If you're using macOS, the command above might not work since macOS doesn't like pip to install libraries globally. Instead, we can create a virtual Python environment by following these steps:

- Navigate to the **"websocket_server"** folder and open a terminal/command line window there
- Run the following command: **"python3 -m venv venv"** or **"python -m venv venv"**
- Now run the following command: **"source venv/bin/activate"**. You should now see **"(venv)"** followed by your current directory (similar to the screenshot)

```
websocket_server — anbr0785@C598 — ..socket_server — -zsh — 80x24
[+] websocket_server source venv/bin/activate
(venv) → websocket_server
```

- Now you may run **"pip install websockets"**
- Test if everything is installed correctly by running **"python WebsocketServer.py"** or **"python3 WebsocketServer.py"**
- If you see something similar to the following screenshot (IP and port can be different), and no errors, you have installed everything properly

```
websocket_server — python WebsocketServer.py — python — Python Web...
[(venv) → websocket_server python WebsocketServer.py
WebSocket server running on IP 10.2.0.2 port 8081
```



Last modified: Wednesday, 4 February 2026, 10:54

[◀ Lab 4 - Tangible Interaction with MQTT](#)

Jump to...

[Lab 5 - Immersive and Tangible Interaction with Websockets ▶](#)